4 Microwaves of the same wavelength and amplitude are emitted in phase from two point sources X and Y, as shown in Fig. 4.1.

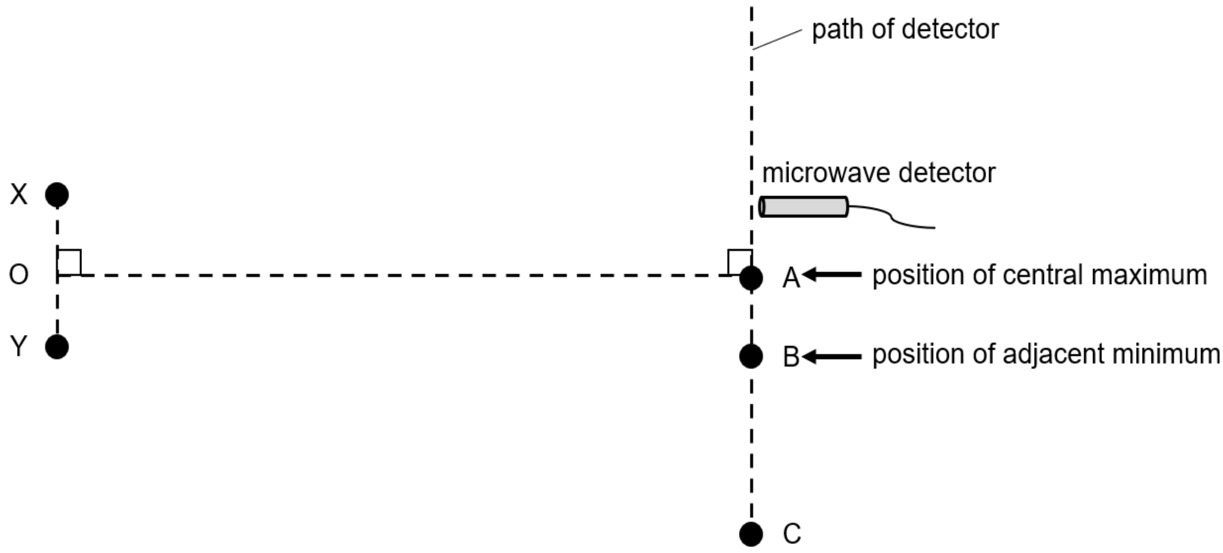

Fig. 4.1 (not to scale)

(a) State and explain along which of the lines XY and OA do the microwaves superpose to produce a stationary wave.

XY. [A1]

Only along XY the microwaves travel in a direction **opposite** to one another. [M1]

Do not accept: "Different direction"

(b) A microwave detector is moved along a line from A to C. The microwave detector gives a maximum intensity reading at A and the first minimum reading at B. The microwaves have a wavelength of 4.0 cm.

For the waves arriving at B, determine the path difference.

B is at position of 1st minima. Hence path difference is 0.5λ.

Path difference = 0.020 m [A1]

path difference = ... m [1]

(c) Describe the effect, if any, on the intensity of the microwave detected at A and B when the following changes are made, separately to the sources X and Y:

(i) when the amplitude of both source X and Y is doubled.

Resultant amplitude is doubled. Intensity is proportional to $(\text{amplitude})^2$. [B1]

Maximum intensity increased by a factor of 4 [B1]

....[2]

(ii) the amplitude of one of the sources is halved.

Resultant amplitude at maxima is 1.5 x_0 and minima at 0.5 x_0. [B1]

Maximum intensity at A becomes 0.5625 or (9/16) times of initial intensity and minimum intensity at B becomes 0.0625 or (1/16) of the initial intensity at A. [B1]

....

....[2]

(iii) the sources are now anti-phase.

the maximum becomes a minimum **and** the minimum becomes a maximum [B1]

the intensity at A and B will swap (B1)

[Total: 8]

5 Fig. 5.1 shows the electric field in the region between two points P and Q. The electric potential